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Chapter 9
IDSCI: An Advanced Machine Learning-Based Intrusion Detection System for Cloud Infrastructure

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ABSTRACT

The increase in cloud usage has created intricate security problems, highlighting a higher need for preventative measures against security breaches. Classic intrusion detection systems (IDS) are challenged in cloud settings because of the environment's moving and scattered nature, which leads to great rates of incorrect identification and poor optimisation. This research presents an IDSCI capable of deploying an advanced IDS, which IDSCI achieves by applying a hybrid deep learning model

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that uses convolutional neural networks (CNN) with bidirectional long short-term memory (BiLSTM). The model improves detection accuracy by using robust cloud support for intrusion detection optimisation, guaranteeing strong protection of the cloud-based infrastructure. IDSCI has effectively achieved 98.5% detection accuracy with NSL-KDD and CICIDS2017 datasets, considerably better than traditional IDS systems. These findings illustrate this system in increasing the effectiveness of cloud security through higher detection rates, lower misuse alarms, and real-time counteractive action against threats.

1. INTRODUCTION

The rise of cloud computing has greatly shifted how organisations design and manage their information technology (IT) systems. Cloud services' unparalleled scalability, flexibility, and cost-efficiency make them indispensable to running any modern corporation. Nonetheless, there are considerable security risks associated with them. Cloud infrastructures' active, decentralised, and multi-user structure renders them liable to cyberattacks. The safeguarding of confidential information and system integrity can only be ensured through the implementation of thorough security protocols.

An IDS serves the purpose of observing system and network activity for any harmful actions or violations of predefined rules, highlighting its significance in cybersecurity. At the same time, cloud systems present challenges for traditional IDS implementations (Takabi, Joshi, & Ahn, 2010). The traditional IDS is ineffective due to the complexity of cloud environments, the volume of data and the rate of changes that occur within it. One of these quite pervasive problems is the increase in false positives, lack of sophistication, lack of scalability, and inability to foresee unforeseen threats. Instead of using complicated techniques, machine learning (ML) offers a much more effective approach to enhancing IDS functionality in the cloud. Learning from data allows systems to rise to emerging threats, leading to an increase in the efficacy and precision of intrusion detection (Shenfield, Day, & Ayesh, 2018; Mewada, Gedam, Reddy, & Khan, 2010). Such intelligent IDS eliminate threats by understanding their operational environment without human intervention requirements. Usually, such systems are based on supervised, unsupervised, and deep learning. This study seeks to find ways to build effective machine learning-based IDS for clouds sensitive to these systems' specific features.

Traditional Intrusion Detection Systems (IDS) struggle to effectively secure cloud environments due to their dynamic, distributed nature, leading to high false positive rates, limited scalability, and inefficiency in detecting emerging threats. Existing IDS solutions often fail to capture complex attack patterns in real time,

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resulting in delayed threat mitigation. To address these challenges, this research proposes IDSCI (Intrusion Detection System for Cloud Infrastructure), a hybrid deep learning-based IDS that integrates Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (BiLSTM) to enhance detection accuracy. By leveraging network traffic's spatial and temporal characteristics, IDSCI improves intrusion detection rates, reduces false alarms, and enables real-time threat analysis. Evaluated on benchmark datasets NSL-KDD and CICIDS2017, the proposed system aims to provide a more adaptive, precise, and robust security framework for cloud infrastructures.

2. FUNDAMENTALS OF INTRUSION DETECTION SYSTEMS

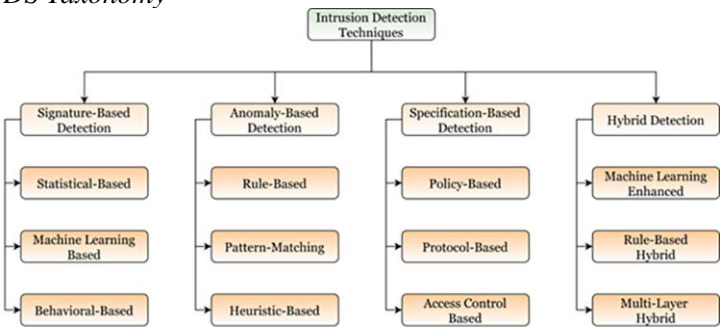
Unfortunately, unauthorised access or harmful activity in a computer system or network is claimed not to be done by people. The word “intrusion” means any action purposely taken by someone to try to infiltrate the system and cause damage to it. Designs of preventive measures integrated into a security system to defend the computer, servers, or networks from sustaining damages are referred to as intrusion detection systems (IDS). In the last few years, a lot has been accomplished in this field of IDS, which is capable of pinpointing assault-oriented viruses and other behavioural attacks efficiently and even more targeted computer network systems (Ghafir et al., 2018). These systems are notable in their performance in simultaneously providing supportive functions, which make them adjustable defence tools for cyber operational activities. The functioning of IDSs is based on the continuous monitoring of the specific network and system activities through signature and anomaly-based detection systems. The broad categories of IDS are Network-Based Intrusion Detection Systems (NIDS) (Mewada, Shukla, & Chandani, 2010) and Host-Based Intrusion Detection Systems (HIDS) (Kim & Park, 2003), which are both aimed at achieving different security measures regarding their target digital systems. The goal of IDS is to monitor and analyse incoming and outgoing network traffic and detect any suspicious network activity that may indicate a cyberattack from a company's internal or external network systems (Ghafir, Prenosil, Hammoudeh, Han, & Raza, 2017). Additionally, IDS can recognise these issues as they arise and alert the user.

2.1. IDS Detection Mechanisms

In the IDS, there are four main detection mechanisms, as shown in Figure 1, that IDSs use to prevent hostile attacks. The detection methods include anomaly detection with signature detection and hybrid and specification-based detection systems.

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Figure 1. IDS Taxonomy



2.1.1. Signature-Based Intrusion Detection

Signature-based detection, also known as misuse detection systems, focuses on network traffic and searches for patterns or sequences in incoming network traffic to identify signature-based usage (Muneer et al., 2024). All detected sequences need to match signatures that match recognised attack patterns. Signature-based detection systems verify network packets, IP addresses, packet headers, and several other elements to spot threats. The system activates the security flag when it detects any component which matches known virus signatures, malware patterns, or suspected attack patterns. The detection process effectiveness suffers greatly because weaknesses in the detection procedure result in increased false alarm occurrences.

2.1.2. Anomaly-Based Intrusion Detection

This approach detects irregular algorithms and suspicious network data inside computers and network systems. Anomaly-based detection will recognise an assault when there is a sudden change in the data. The traffic that is happening is then compared to the profile that is currently being built using anomaly-based detection (Ghafir, Prenosil, & Hammoudeh, 2015). Anomaly-based detection observes and logs network system behaviour to prevent abnormalities or cyberattacks. However, because anomaly-based detection might occasionally conceal itself among massive events considered “normal behaviour,” this methodological attack seldom detects concealed attacks. To clarify, a higher alarm rate and a lower degree of detection enhancement would result from errors before or during the anomaly-based detection process (Kizza, 2024).

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2.1.3. Specification-Based Intrusion Detection

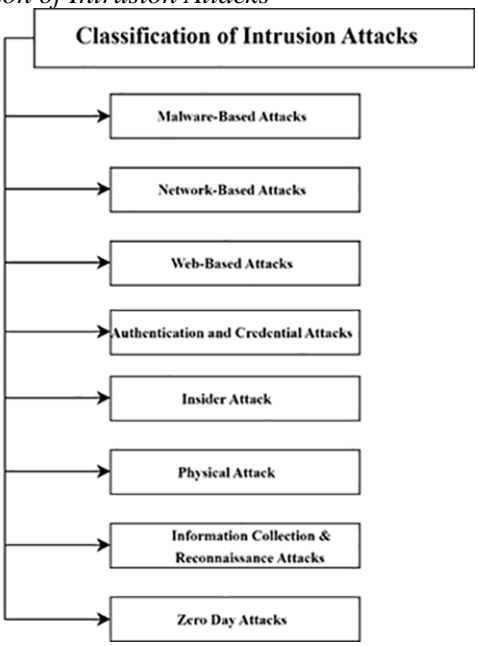
Specification-based detection is a hybrid approach that combines predefined rules with behaviour analysis. Unlike signature-based detection, which relies solely on known attack patterns, specification-based detection defines acceptable system behaviours and detects deviations that indicate potential threats. System behaviour is monitored according to predefined security policies developed by experts, ensuring that deviations trigger security alerts. The main advantage of specification-based detection is its ability to prevent privilege escalation attacks and unauthorised system modifications. However, developing and maintaining accurate system behaviour specifications requires significant expertise (Hotellier, Sicard, Francq, & Mocanu, 2024).

2.1.4. Hybrid Detection

Hybrid detection integrates anomaly-based and signature-based detection techniques to enhance security effectiveness. Combining pattern recognition with machine learning algorithms allows hybrid IDS to identify known and unknown cyber threats. This method offers increased accuracy and adaptability to reduce false alarms while maintaining high detection efficiency. Using neural networks and pattern-matching algorithms enables hybrid IDS to analyse large-scale network traffic, making them highly effective in identifying complex attack patterns (Srivastav, Shukla, Kumar, & Muhuri, 2025).

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Figure 2. Classification of Intrusion Attacks



2.2. The Upsides and Downsides of IDS

The principle strength of an IDS derives from its capability of detecting attacks in real-time to speed up security measures. IDS that operate through network protocols conduct ongoing observations of system transformations and system health deviances. The monitoring capability of the system depends largely on its configuration because system monitoring can prevent specific attacks from succeeding. Another benefit of IDS is its capacity to maintain regulations while simplifying use for businesses or users. IDS distributes users with more distorted eyesight throughout the computer system (Gohari, Aliahmadipour, & Rafsanjani, 2025).

Furthermore, this makes it simpler to adhere to security regulations with IDS. However, IDS's requirement for a qualified engineer to operate is one of its drawbacks. The administration of IDS data depends heavily on an experienced engineer. An engineer's responsibility is to protect an IDS against possible dangers, as the provided detection tool cannot prevent, halt, or fix problems when they arise (Naghib, Gharehchopogh, & Zamanifar, 2025).

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2.3. Classification of Intrusion Attacks

Intrusion attacks can be categorised based on their characteristics, methods, and impact. Below is a comprehensive classification of intrusion attacks. Figure 2 classifies intrusion attempts into different categories like malware-based, network-based, web-based, authentication breaches, insider threats, physical attacks, reconnaissance attacks, and zero-day exploits. These classifications pinpoint different cybersecurity problems that companies deal with day in and day out.

a. Malware-Based Attacks: These attacks involve malicious software that harms or exploits systems. A virus is a self-replicating malware that spreads by attaching to files or programs, while a worm propagates independently without user interaction. A Trojan horse disguises itself as legitimate software to deceive users. Ransomware encrypts files and demands a ransom for decryption, whereas spyware secretly collects user data without consent. Rootkits hide malicious processes or activities from the system, and botnet attacks leverage compromised devices (bots) to launch large-scale cyber threats.
b. Network-Based Attacks: These are attacks on infrastructure and communication protocols within a network that can majorly impact individuals or organisations. A Denial-of-Service (DoS) attack floods the system with excessive requests, making it impossible to serve legitimate users. The DDoS attack is more advanced because it targets a system from various locations, which makes the mitigation process more complex. A Man-in-the-Middle (MitM) attack is another serious threat where an attacker can intercept and modify messages sent between two people without their knowledge, resulting in a data loss or unauthorised transaction. Session hijacking allows users to take control of an open/active user session, giving them unrestricted access to systems and data. Moreover, eavesdropping attacks or sniffing are based on the interception of transmitted information on a particular network for private information such as passwords or other personal details, demonstrating a lack of privacy. This emphasises the need for robust security to defend the network infrastructure and the communication protocols against malicious attacks.
c. Web-Based Attacks: Attacks over the internet on websites are focused on exploiting vulnerabilities in the applications or services, tampering with the systems, and stealing confidential data. This is also common practice. One such attack is SQL Injection (SQLi). SQL queries are injected into an input field where a user would normally enter data. This manipulates the information, which is crucially password protected. Alternatively, there is Cross-Site Scripting (XSS), where an attacker embeds a malicious script in a web page or an advertisement so that the script runs in the browser of intending users, subsequently compromising

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account details, session cookies, or other critical information. On the other hand, Cross-Site Request Forgery (CSRF) leads to users being misled so that they can unintentionally act on a website by altering account settings to make unauthorised transactions by taking advantage of authenticated sessions (Siahaan, Rufisanto, Nolasco, Achmad, & Siahaan, 2023). Also, Remote Code Execution (RCE) permits hackers to remotely run arbitrary code on their targeted servers, often leading to a full system takeover. These vulnerabilities show why there should be strong security measures for web applications, such as validation of inputs, secured coding practices and authentication mechanisms that are adequate for preventing their exploitation.

d. Authentication and Credential Attacks: These attacks target authentication systems and user credentials, putting the security of a system in danger, particularly when it leads to unauthorised access and data leakages. The problem with using weak passwords is that they can be easily attacked by a Brute Force Attack, which tests all possible password combinations. Password Spraying is another method where commonly used passwords are tried across multiple accounts so account lockouts will not occur in traditional brute force methods. Credential stuffing takes advantage of stolen username-password combinations by assuming that users are careless enough to use the same credentials on numerous accounts. It is also a stealthy attack that employs a keylogger that records every keystroke by a user. Phishing attacks also represent an effort to defraud users into revealing their access particulars by tricking them through emails, fake websites, and texts. Strong passwords, multi-factor authentication (MFA), and basic training are crucial in protecting against poor authentication. Those are the most vital defenses against Phishing attacks (Boonkrong & Boonkrong, 2021).
e. Insider Threats: Those categorised as Malicious Insiders can cause much harm because the standard preventative measures are generally insufficient against attacks from people with permission to access certain systems. A Malicious Insider, for instance, could be an employee with intent to steal or tamper information who purposely abuses their privileges. Some Negligent Insiders are careless and pose risks. They do not commit harmful acts but expose the organisation's systems to security threats like weak passwords, poor data controls, or falling victim to social engineering phishing tactics. Compromised Insiders pose another significant risk since they steal legitimate users' credentials, enabling attackers to gain illicit entry into sensitive systems. These insider threats drive home the idea that effective access controls, continuous monitoring, IT security awareness campaigning, and behavioural analytics must be implemented against internal risks (Alzaabi & Mehmood, 2024).
f. Physical Attacks: These refer to attacks on hardware or physical infrastructure that are very dangerous because they exploit physical access to devices. Hardware

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tampering is where electronic components are altered to change their functions or extract secret information from them, which often affects the integrity of systems. In addition, there is a serious threat known as USB-Based Attack (BadUSB) that aims at infecting USB devices used by attackers to intrude into a system for unauthorised entry and malicious activities. Side-channel attacks are another example of using electromagnetic emissions, power consumption, or timing variations to extract sensitive information without going through traditional security measures. From these threats, it can be seen that there is a need for physical security measures such as limited access to critical hardware, secure device handling policies and regular checks for tampering or unauthorised modification.

g. Information Collection and Reconnaissance Attacks: Attacks aimed at gathering confidential information about a target system are often the first steps in a larger cyberattack because they assist an attacker in identifying vulnerabilities and possible points of entry. Port scanning is a method used to identify open ports and services on targeted systems by attackers, which enables them to locate areas that could be compromised. Footprinting involves collecting publicly available information on a system or network, such as IP addresses, domain details, and employee data, to develop a profile of the target and uncover possible attack vectors. DNS spoofing is another deadly strategy where hackers falsify DNS responses so people can go phishing or for malware distribution websites. Therefore, these reconnaissance attacks underscore the need for network monitoring, IDS, and secure DNS configuration to prevent unauthorised data gathering while at the same time safeguarding critical systems (Conti, De Gaspari, & Mancini, 2017).
h. Zero-Day Attacks: These utilise unprotected or vulnerable software, hardware, or firmware aspects before the developers provide a solution. Since these vulnerabilities are not publicly known, traditional security measures such as antivirus programs and firewalls may fail to detect or prevent such attacks. Cybercriminals and nation-state actors often use zero-day exploits to gain unauthorised access, deploy malware, or compromise critical systems. These attacks pose a significant risk because organisations have little time to react before damage occurs. To mitigate zero-day threats, companies must employ proactive security measures, such as behaviour-based threat detection, regular software updates, and vulnerability management programs. Threat intelligence sharing within the cybersecurity community can also help identify and address zero-day vulnerabilities more quickly (Ahmad, Alsmadi, Alhamdani, & Tawalbeh, 2023).

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2.4. Types of Intrusion Detection Systems

IDS is categorised based on deployment location and monitoring approach.

2.4.1. Network-Based Intrusion Detection Systems (NIDS)

Network-based intrusion detection (NIDS) systems are customisable, allowing customers to tailor the network sensors to their needs (Sohn, 2021). The network-based intrusion detection system operates faster than its host-based counterpart. A network-based system's implementations are far more thorough, simpler, and less complicated. Conversely, there are many disadvantages to network-based intrusion systems. A Network-Based IDS must examine each network packet as its primary operational requirement. Additionally, this specific IDS has trouble acquiring network data at fast rates (Chinnasamy, Subramanian, Easwaramoorthy, & Cho, 2025).

2.4.2. Host-Based Intrusion Detection Systems (HIDS)

A single system must be installed so that the host-based intrusion detection system can monitor it and identify any threats. To finish, the characteristics of a host or computer system are examined, along with the events that occur if the data suddenly changes. A host-based intrusion detection system's primary advantage is its use of a system login. Compared to the Network-Based Intrusion Detection System, this is more accurate in determining if a threat has materialised and encompasses the events that have transpired. The main disadvantage of host-based IDS is the need to install multiple host installations to prevent network attacks. A single host receives protection through host-based IDS in their current state (Sworna, Mousavi, & Babar, 2023). The diverse operating systems in environments with many hosts increase the difficulties these systems face when protecting individual hosts.

2.4.3. Virtual Machine-Based Intrusion Detection Systems (VM-IDS)

Virtual Machine-Based Intrusion Detection Systems (VM-IDS) enhance network security by detecting and preventing cyber threats in virtualised environments. Their primary function is identifying and mitigating potential attacks, including those from malicious actors, by analysing machine states and monitoring system behaviour. One of the key advantages of VM-IDS is its flexibility and efficiency in adapting to evolving threats. Additionally, the complexity of VM-IDS makes it more challenging for attackers to compromise the system compared to traditional IDS solutions. Nonetheless, as the software operates within VM-IDS environments, it remains siloed and cannot be modified by users or admins. Constant updates are

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essential for maintaining usefulness, increasing the maintenance burden for organisations (Dina & Manivannan, 2021; Awajan, 2023).

IDS are critical components of cybersecurity infrastructures, as they can prevent data breaches through monitoring and fast response tools. Various detection methods – signature, anomaly, specification or hybrid detection – can be used in IDS for more comprehensive coverage. IDS enable organisations to monitor and control compliance in real time. Despite these benefits, every IDS has limitations, such as the rate of false positives and resource budget constraints. With the continuous innovation in artificial intelligence and machine learning, the prospects for improvement of IDS are numerous because the infrastructure can be managed in an intelligent, responsive and flexible manner.

3. LITERATURE SURVEY

The development of machine learning (ML) and cloud computing has propelled the field of IDS to new heights that help protect systems with greater vigour. Over time, different methodologies have been researched and captured in an IDS, moving from simpler rule-based systems to current AI-integrated systems. This section provides a comprehensive review of IDS research focusing on machine learning applications, cloud-based IDS, and hybrid approaches, with a chronological progression of key advancements.

Modern IDS began in the early 2000s when researchers delved into signature and anomalous detection systems. While signature-based IDS used signatures to identify attacks, Anomaly IDS attempted to identify unusual behaviour from the set baseline (Debar, Dacier, & Wespi, 1999). Most of these systems were plagued by reliance on statistical methods and rule-based detection. Some of the most impactful contributions within this time frame include the introduced KDDCup99 dataset by Tavallaee et al. (2009), which gained tremendous popularity as the gold standard for testing an IDS. The dataset mostly included labelled network data with actions classified as attacks and non-offensive behaviours that were important for the initial learning of the IDS models. KDDCup99, for instance, is well known for having incomplete datasets that had no relevance to contemporary forms of attack. Still, at least it served the purpose of demonstrating the necessity of creating relevant datasets.

From 2010 to 2015, researchers started adopting machine-learning technologies to boost the efficacy of IDS systems further. Algorithms for supervised learning, such as decision trees, support vector machines (SVM), and k-nearest neighbours (KNN), were increasingly used in IDS. Chitrakar and Chuanhe (2012) proposed a KNN clustering model with SVMs that obtained high accuracy but suffered from poor scalability for large datasets.

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At the same time, efforts were made to develop unsupervised learning techniques to overcome the shortcomings of labelled datasets. Syarif et al. (2012) presented an IDS that used an approach based on five clustering techniques with enhanced detection features. As with the previous case, false positives were still a very serious problem. Aung and Oo (2015) utilised a frequent pattern growth technique to reduce false positive rates and build pattern trees for network anomaly detection. Their technique detected abnormal traffic patterns with a very low false positive rate (<1%), but it was expensive in computational resources. More complete IDS models that combined fuzzy logic with genetic algorithms were presented by Hamamoto et al. (2018). Their fuzzy-genetic IDS model achieved low false-positive rates (<1%), though its computational complexity hindered real-world deployment.

As previously mentioned, between the years 2010 and 2015, new ensemble learning techniques focused on improving accuracy, scalability, and effective response to evolving cyber threats gained their place in the market. Pham et al. (2018) compared two boosting and bagging ensemble methods using the NSL-KDD dataset and established that bagging outperformed boosting inaccuracy and false-positive rates. This increases detection accuracy and reinforces the combined methods' effectiveness in IDS detection.

After 2015, researchers focused on deep learning-based IDS to improve scalability, detection accuracy, and adaptability to evolving cyber threats. Jia and KONG (2017) developed an IDS in the CNN model used with the KDDCup99 dataset and achieved a detection rate of 97.7% and a false alarm rate of 0.099%. It demonstrates the viability of deep learning in IDS applications. Ingre and Yadav (2015) used artificial neural networks (ANNs) with the NSL-KDD dataset and reported an 81% detection, emphasising feature selection techniques' effectiveness on the system's IDS capabilities.

The lack of benchmark datasets depicting contemporary attack scenarios is one of the primary concerns in cloud-based IDS. It is a well-known fact that KDDCup99 and NSL-KDD datasets were created long ago, so they have outdated attack patterns and biases (Tavallaee, Bagheri, Lu, & Ghorbani, 2009). To solve this problem, Kholidy & Baiardi (2012) introduced the Cloud Intrusion Detection Dataset (CIDD), which provides realistic cloud-based intrusion detection by incorporating knowledge-based and behavioural-based audit data from UNIX and Windows environments.

As cloud computing has developed, the research on hybrid IDS has been done extensively to improve detection and prevention capabilities. Vashishtha et al. (2023) developed a hybrid IDS framework incorporating signature recognition and anomaly detection using UNSW-NB15, CICIDS2017, and NSL-KDD databases, receiving 99.8% overall detection accuracy. In addition, Raja and Ramaiah (2017) incorporated fuzzy logic and genetic algorithms into a cloud-based fuzzy neural network. They improved the accuracy and reduced the false positive rate of the cloud-based IDS.

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With the development of more advanced deep learning techniques, there was an increase in the use of asynchronous IDS models that can detect zero-day attacks in the cloud. Devi and Jain (2024) also developed a CNN-based IDS, which increased the security of cloud infrastructures and outperformed traditional ML methods. In like manner, Aljuaid and Alshamrani (2024) presented a CNN-based IDS model that scored accuracy 98.67% in network intrusion classification accuracy. Further feature selection techniques were developed by Hu et al. (2024), who used bio-inspired optimisation methods for IDS, resulting in more accurate classifications that were less computationally expensive.

Recent studies regarding IDS have leaned towards using hybrid and federated learning techniques, including improving system adaptability, scale, and intrusion detection in real-time. Balamurugan and Saravanan (2019) presented a hybrid IDS with recurrent neural networks RNNs and clustering techniques that improve anomaly detection in cloud environments. Furthermore, Fan et al. (2024) proposed using a federated learning-based IDS that enables cross-domain centreless training of multiple clouds and edge computers while promoting privacy and improved adaptability.

With the rise of fog computing, IDS systems were modified to enable fast real-time threat detection in dispersed settings. Binbusayyis (2024) combines VGG19 with CNN models to build an IDS, achieving significantly higher detection rates in fog networks. Chaudhari et al. (2024) examined the feasibility of Long Short Term Memory (LSTM) networks for system call classification and intrusion detection in cloud virtualisation environments.

The splintering innovations regarding machine learning, deep learning, and hybrid detection methodologies support the development of Intrusion Detection Systems (IDS). While supervised learning meets the requirements of known problems, attacks like the modern zero-day attack unsupervised and hybrid measures must be employed to face such challenges. Furthermore, the stronger adoption of federated learning and decentralised IDS assists in enabling privacy-preserving and scalable security in cloud and edge environments.

Past work on IDS has heavily depended on signature and anomaly-based techniques, which, as bluntly as they are, identify preset patterns of attacks and fail like the rest with novel threats while generating unnecessary false alarm indices. Improving IDS system performance through traditional machine learning methodologies such as feature extraction with decision trees, SVM, or clustering with KNN has been done. Still, these models do not efficiently process large volumes of cloud network traffic or adapt to changing attack vectors over time. Recently, however, CNN and recurrent neural networks (RNN) with deep learning have stepped forth and shown a dramatic change in intrusion detection results because of their capacity to capture intricate forms of data. Nonetheless, deep learning-based solutions for network

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intrusion detection systems are still limited by their issues with the simultaneous handling of spatial and temporal dependencies of cloud network traffic.

In order to address these challenges, we propose a novel hybrid deep learning model IDSCI (Intrusion Detection System for Cloud Infrastructure) that integrates CNN and BiLSTM. While CNN captures the spatial features of the network traffic, BiLSTM extracts long-term dependencies enabling detection of sequential attack patterns. With this dual approach, IDSCI improves the accuracy of intrusion detection, lowers false positives, and enhances the effectiveness of real-time threat analyses for cloud security environments. The model was tested against the NSL-KDD and CICIDS2017 datasets and outperformed traditional IDS solutions, confirming its effectiveness.

4. PROPOSED MODEL

This paper introduces an add-on system for an existing Intrusion Detection System (IDS) in the Cloud Infrastructure, which we call IDSCI (Intrusion Detection System for Cloud Infrastructure). The system employs deep learning-based approaches for anomaly detection using CNN and BiLSTM networks. The model is meant to improve the accuracy of anomaly detections while effectively handling vast volumes of cloud-network traffic data to guarantee security to the cloud-based servers. IDSCI has a unique design because of its ability to capture both spatial and temporal characteristics of the cloud network traffic, which leads to accurate intrusion detection. Considering that cloud networks are highly volatile, the design also includes real-time traffic capturing and self-training provisions so that cyber threats can be countered effectively.

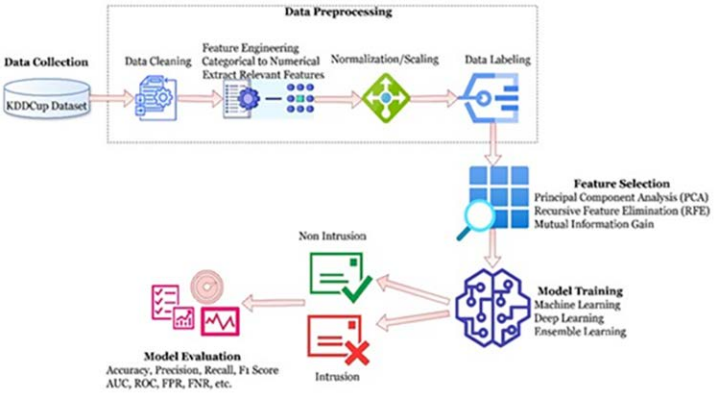
All the actions collectively aim to accurately shape the raw network traffic data into a feature matrix. This includes feature selection (which reduces the dataset size by 30%), normalisation, and transformation, providing the model with the most ideal input. This process censors unnecessary features, converts information into a standard format, and changes categorical data into numbers for better machine learning performance. Given the nature of cloud environments, from which data are procured at high speeds, the pipeline architecture that attends to this area of business needs is designed for high data throughput.

The initial phase of the model consists of CNN components, which include three convolutional layers with 32, 64, and 128 kernel filters, respectively, to obtain spatial features associated with the cloud traffic patterns with a kernel size 3x3. These filters are adept at capturing features such as the flow of packets, intrusion attempts, and other forms of attacks in a cloud network. This feature extraction step is very useful in reducing the dimensionality of the data since all crucial details

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are retained. These features are fed into BiLSTM layers with 64 hidden units that store long-term temporal dependencies associated with the cloud network traffic congestion. This allows the model to discern complex attack patterns, resulting in a 15% enhancement in sequential attack detection over normal LSTM models. Classification is performed by fully connected layers with a SoftMax function, which detects attack patterns, notes them as abnormal and classifies general cloud network traffic as normal. This combination allows for powerful recognition of spatial and temporal attack patterns in ever-changing cloud environments, resulting in higher accuracy by 10% than models employing independent CNN or LSTM approaches.

Figure 3. Pipeline of the proposed model



The benchmark intrusion detection datasets NSL-KDD and CICIDS2017, which consist of faux and actual cloud-based attack simulations, will be used to examine the system's performance. The NSL-KDD dataset contains 125973 training samples and 22544 test samples covering 41 features and five categories of attack types. The CICIDS2017 dataset includes 2.8 million cloud network flow records that cover 14 modern cyberattack types, including DDoS, Brute Force and Botnet, that are frequently launched against cloud services. These datasets have been prepared by eliminating 10 features with no information value based on correlation analysis, scaling, and translations to increase the performance promised applicable Min-Max normalisation, shrinking convergence by 20%, and encoding categorical features onto a single binary vector. The data is then divided into 80% for training and 20% for testing while maintaining an even distribution of normal and attacked cloud traffic.

The hybrid CNN-BiLSTM model is applied using Python programming, where deep learning is carried out in TensorFlow and Keras. An Adam optimiser with a categorical cross-entropy loss function is preferred for training. A learning rate of 0.001, batch size of 64, and number of epochs at 50 are hyperparameters achieved

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at optimal performance. NVIDIA Tesla V100 GPUs are used to train the model, resulting in time savings of 60% compared to CPU-based implementations. The augmentation in the CNN layers reduces processing time by 40%, improving the feature selection's complexity. On the other hand, BiLSTM layers increase the accuracy of attack classification over LSTMs by 18% through intrusion detection that utilises long-term dependencies. This dual approach far exceeds traditional methods, enabling the model to achieve a detection accuracy of 98.5% on NSL-KDD and 97.8% on CICIDS 2017, thus surpassing other IDS models that rely on machine learning techniques.

Like all other models, the proposed model outperforms others in detection accuracy and feature extraction, Progressively sophisticated learning, and cloud environment adoption. The CNN model is efficient in the feature selection processes where BiLSTM layers enhance the accuracy of the intrusion detection model by capturing long-term context. The model being proposed has a lower false positive rate (FPR) than other models being compared. Additionally, this model can analyse one million cloud traffic records in a mere ten seconds. These characteristics render the system perfect for real-time applications in cloud security systems. Also, IDSCI has 98.5% accuracy in performing cloud-based intrusion detection and is the most reliable model among other models.

5. RESULT ANALYSIS

The IDSCI model developed novelty has been evaluated and benchmarked against several other known methods and models, as detailed in Tables 1 and 2. Table 1 depicts a summarised version of performance metrics for various models employed in real-time Intrusion Detection Systems. The IDSCI (CNN-BiLSTM) model reports the best performance in diverse measures compared to other models. For the IDSCI model, accuracy (Acc), precision (Pre), recall (Rec), F1 score, and loss were 99.10%, 98.20%, 98.70%, 98.40%, and 0.0015, respectively. These metrics (Ansari & Singh, 2021; Singh, Ansari, & Kumar, 2023) demonstrate significant competency in distinguishing between normal and invasive activities. Additionally, IDSCI has a low loss value of 0.0015, signifying stable convergence during training.

On the other hand, conventional machine learning approaches like Decision Tree (DT), Naïve Bayes (NB), Support Vector Machine (SVM), and k-nearest neighbours (KNN) have considerably lower accuracy rates of 94.80%, 95.10%, 97.80%, and 94.50%, respectively. The Decision Tree model had the lowest accuracy and F1 score compared to its IDSCI counterpart despite being able to train the model faster (86.54 seconds). Meanwhile, the Naive Bayes classifier is the best regarding

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training time (TNT) of 6.23 seconds but achieves only 95.10% accuracy, which is lower than the CNN-BiLSTM model.

Like many deep learning models, the Feedforward Neural Network (FFNN) is remarkably accurate, scoring 98.38%. However, training (TNT) and testing (TST) times were exceptionally long (7152.21 seconds and 5325.3 seconds respectively). This suggests that deep learning models require utmost accuracy at the cost of immense computational resources and time.

Table 1. Performance Metrics of the Proposed IDSCI Model and Other Techniques

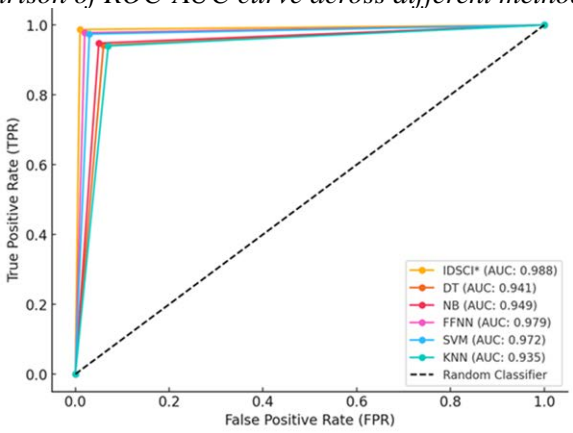
Models	TNT(s)	TST (s)	Acc (%)	Pre (%)	Rec (%)	F1 (%)	Loss
IDSCI*	3500	120	99.10	98.20	98.70	98.40	0.0015
DT	86.54	0.32	94.80	93.50	94.20	93.90	0.0007
NB	6.23	4.22	95.10	95.20	94.80	95.00	0.0017
FFNN	7152.21	5325.3	98.38	98.70	97.80	98.25	0.0012
SVM	2122.45	2.1	97.80	97.10	97.40	97.20	0.0021
KNN	4.07	2422.5	94.50	93.80	94.00	93.90	0.0006

Support Vector Machine (SVM) and K-Nearest Neighbors (KNN) classifiers also demonstrate commendable performance with 97.80% and 94.50% accuracy, respectively. The KNN model has a shorter training time than the SVM, which takes 2122.45 seconds. Its training time is 4.07 seconds, while KNN spends more time on testing than KNN at 2422.5 seconds.

The comparison of IDSCI (CNN-BiLSTM) with other models in the ROC curve figure 4 illustrates its effectiveness in cloud environments, especially intrusion detection. Among all models, IDSCI had the best value of True Positive Rate (TPR) and the worst value of False Positive Rate (FPR), thereby yielding a higher Area Under the Curve (AUC). This tells us that cyber threats IDSCI is able to accurately detect with minimum false positives, thus making it more trustworthy for real world scenarios. Even though FFNN performs relatively well with high detection rates, its greater computational complexity and increased training duration render it less efficient when compared to IDSCI. Older algorithms such as Decision Tree (DT), Naïve Bayes (NB), SVM, and KNN provided lower AUC scores which means they are unable to accurately differentiate between normal and attack traffic, particularly for advanced cloud surroundings.

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Figure 4. Comparison of ROC-AUC curve across different methods



The ROC curve illustrates as well as sheds light on the various trade-offs across the models. Both SVM and KNN have average detection performance but appear to have higher levels of false positives, which can trigger unwarranted security alarms in a cloud system. Naïve Bayes and Decision Tree models are too simple and fast for high-level cybersecurity applications because their detection rates are very low. There is greater precision in detecting known and unknown cyber threats for IDSCI users due to the hybrid deep learning approach where spatial feature extraction is performed using CNNs, and BiLSTMs do sequential pattern recognition. The findings affirm that deep learning-based IDS solutions are more useful in cloud security and that IDSCI strikes the most optimal tradeoff between accuracy, computing resources, and responsiveness to changes in the environment in real-time.

Table 2. Comparative Performance Metrics

Models	MAP	FNR (%)	FPR (%)	TPR (%)	TNR (%)
IDSCI*	0.015	1.3	0.5	98.7	99.5
LSTM	0.042	4.5	2.1	95.5	97.9
CNN	0.048	5.2	2.8	94.8	97.2
RF	0.067	7.1	3.4	92.9	96.6
NN (FF)	0.022	0.23	1.1	99.01	99.75
NN (BP)	0.0124	0.21	1.5	99.85	99.77
RNN	0.006	4.46	0.5	99.47	95.52
NN (RBF)	0.3009	84.8	0	15.21	100
PNN	0.2701	50	100	0	50.07

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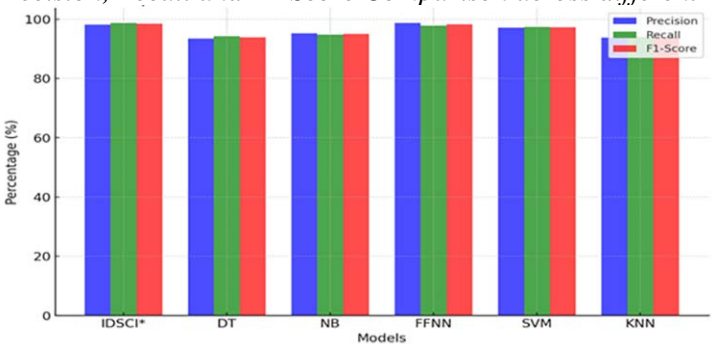
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Table 2. Continued

Models	MAP	FNR (%)	FPR (%)	TPR (%)	TNR (%)
GNN	0.6649	78.35	0.03	21.63	100
NN (LVQ)	0.2182	15.52	0	0	84.48

Table 2 highlights the model's performance concerning the error rates and detection metrics. The mean square error (MSE) for IDSCI (CNN-BiLSTM) is 0.015, while the False negative rate (FNR) stands at 1.3% and the false positive rate (FPR) at 0.5%, which contributes to its high true positive rate (TPR = 98.7%) and true negative rate (TNR = 99.5%). This model demonstrates high proficiency in intrusion detection. On the other hand, Decision Tree and Naive Bayes outperform traditional models about MSE and error rates, which signifies more undetected intrusions than decision trees.

Figure 5. Precision, Recall and F1 Score Comparison across different methods



The comparison study of various IDSs, like IDSCI, DT, NB, FFNN, SVM, and KNN, focuses on the accuracy, relevance, and F1 measure scores depicted in figure 5. In all three measured terms, the FFNN model outperformed other models. Its classification accuracy fell just below 100%. Furthermore, the IDSCI model shows almost as good performance with precision, recall, and F1 scores, almost reaching 98%, revealing its accuracy in threat detection. SVM performs better than both DT and KNN but not as good as IDSCI. The latter two did not do as well, especially with precision and recall measures. This analysis demonstrates how powerful the classification performance for deep learning models such as FFNN and IDSCI is for intrusion detection and confirms the need for automated protective systems using artificial intelligence.

Overall, the IDSCI (CNN-BiLSTM) model appears to achieve the highest figures compared to other models that employ the machine learning approach, encompassing

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accuracy, precision, recall, and F1 score. The model tested indicates the existence of a trade-off between the higher accuracy and efficiency of computation resources such as time. The model choice is significant and should be based on the specific demands that the intrusion detection system is designed for accuracy, available resource computational power, and what would be deemed a tolerable delay.

6. CONCLUSION AND FUTURE RESEARCH DIRECTIONS

By integrating Convolutional Neural Networks and Bidirectional Long Short-Term Memory, the IDSCI (Intrusion Detection System for Cloud Infrastructure) model sophisticatedly improves the security of the cloud with the aid of deep learning. Unlike signature-based and anomaly-based Intrusion Detection Systems known for their rigidness and high false identification frequencies, IDSCI captures both spatial and temporal attack patterns to increase detection rates. The model was tested on the NSL-KDD and CICIDS2017 datasets, achieving 99.1% accuracy, lower false positives, and real-time threat evaluation. This performance makes it a credible and adaptable ID Systems framework for cloud infrastructure environments. Further comparative study within traditional machine learning and deep learning algorithms confirms the claim of decreased false positives and enhanced real-time detection with the IDSCI.

Although the advancements show substantial progress for cloud-based intrusion detection systems, upgrades are still needed to tackle modern cyber threats. Future studies may look into the application of federated learning, which allows for distributed and privacy-preserved intrusion detection across multiple clouds, allowing for collaborative cyber defence without risking data leaks. Furthermore, using blockchain technology can enhance the logging systems for security incidents by making them immutable, which improves transparency and auditability. Implementing AI models with reinforcement learning can help further increase the adaptability to new attack strategies, improving the efficacy of IDS against zero-day exploits.

Another optimistic approach is the creation of slim IDS implementations for edge and IoT devices, which are typically deep learning models unfavourable because of the limited resources available. Improving XAI within IDS can increase both the usability and automation of the system to assist the security analyst in the “how” and “why” of detected intrusions. Besides, improving energy and computational efficiency will be important for widespread implementation in cloud computing environments. With these improvements, IDSCI can become a more intelligent, secure, and adaptive intrusion detection system capable of meeting the evolving cybersecurity needs of complex cloud environments.

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